Retrieval Augmented Generation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Retrieval-Augmented Generation

> Prepare Retrieval Augmented Generation: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of building a **grounded answer**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Model Memory vs Retrieval

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of retrieval and indexing

Reference example, not the only acceptable wording: Teacher explanation: Retrieval returns stored records selected for a **query**.

An index is a searchable representation of a versioned corpus; generated wording is a later operation. Assign stable document IDs, versions and permitted access before indexing. Retrieve only authorised records.

Preserve source references and define behaviour when a document is missing, outdated or access-restricted.

Source Quality

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of conflicting and time-sensitive sources

Reference example, not the only acceptable wording: Teacher explanation: Changing facts require a stated time and source-version policy. Current-looking records can conflict without enough **evidence** to choose between them.

For this toy task a signed amendment supersedes the referenced original from its effective time. Keep both records. A newer timestamp alone has no authority unless the policy grants it.

Speech and Audio Systems

Without opening the old lesson, explain one key idea from Speech and Audio Systems. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**

Reference example, not the only acceptable wording: Teacher explanation: A multimodal pipeline may transcribe speech, inspect an image and generate a response.

Errors at one stage can propagate even when later stages run correctly. Track source modality, segment/time references, transcription, interpretation and final claims separately.

Compare conflicting **evidence** without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> Sources: A='Library opens 9'; B='Cafe opens 8'; C='Library closes 5'. A question asks when the library opens. The **retriever** returns only B and C. What should the generator do?

Concept to explain

Week 18: Retrieval-Augmented Generation

Term	Meaning
query	A question or search request used to obtain information.
retriever	The component that searches for and selects source material relevant to a query.
relevance	How directly information helps answer the stated question or perform the task.
grounded answer	An answer whose factual claims are supported by identified evidence relevant to the question.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. A **grounded answer** connects each factual claim to **evidence** actually retrieved for the task. Retrieval success does not automatically make generation faithful.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Apply the connected idea: Chunking and document structure

Two 45-minute sessions. **Chunk**ing divides documents into retrievable units. A useful **chunk** retains enough structure to interpret a passage, including headers and exceptions.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Apply the connected idea: Lexical and vector retrieval

Two 45-minute sessions. Lexical retrieval matches terms; **vector retrieval** compares numerical representations. Each can fail when the representation misses the relevant distinction.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can explain retrieval augmented generation using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Retrieval Augmented Generation using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Question is converted into a search **query**.

2. **Retriever** returns four passages; selector keeps two relevant ones.

3. Generator answers using those passages and attaches citations.

4. If retrieval misses the needed **evidence**, generation should ask, search again, or abstain.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Note A says library opens at 9. Note B says returns go in the blue box.

The teaching case

Answer opening time with **evidence**.

Exact explanation to model

Why the result follows

Answer 9 with note A. Retrieval supplies **evidence** and generation phrases it. B does not support opening time.

Explain the added worked example

A **grounded answer** connects each factual claim to **evidence** actually retrieved for the task. Retrieval success does not automatically make generation faithful.

Explain the added worked example · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Build a claim-**evidence table** before final wording. Preserve source versions and **qualifier**s, omit unsupported claims or explicitly state the gap, then validate the answer against the table.

Explain the added worked example · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Grounded claim	A claim supported by the available task evidence
Qualifier	A condition restricting a statement
Evidence table	A mapping from each claim to its supporting record

Apply the connected idea: Chunking and document structure

Chunking divides documents into retrievable units. A useful **chunk** retains enough structure to interpret a passage, including headers and exceptions.

Apply the connected idea: Chunking and document structure · 3

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Give each **chunk** its document/version ID, section and **offset**s. Preserve **qualifier**s or link adjacent material. **Overlap** can repeat information; repeated **chunk**s are not independent sources.

Apply the connected idea: Chunking and document structure · 4

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Apply the connected idea: Chunking and document structure · 5

Guided: The first **chunk** alone lacks the applicable exception. Retrieve the linked condition and determine maintenance status before asserting capacity 20.

Qualification: Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Apply the connected idea: Lexical and vector retrieval

Lexical retrieval matches terms; **vector retrieval** compares numerical representations. Each can fail when the representation misses the relevant distinction.

Apply the connected idea: Lexical and vector retrieval · 6

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

For a toy lexical score, count distinct **query** terms present after lowercase exact-word matching. For a toy vector score use a stated dot product. Compare **relevance** using independent judgments, not score magnitude alone.

Apply the connected idea: Lexical and vector retrieval · 7

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Apply the connected idea: Lexical and vector retrieval · 8

Guided: Lexical ranks D1 first; dot product ranks D2 first (.8 versus .4). The supplied task **relevance** favours the sensor manual; vector ranking alone does not prove **relevance**.

Qualification: Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch.

Approval and capacity are supported by P and Q; lunch lacks **evidence**. A **grounded answer** reports the first two claims and does not invent lunch.

Model the reasoning aloud · 9

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Retrieval Augmented Generation

Turn a short science source into an answer with a visible claim-to-source map.

The facts we will use

Supplied facts

Approved note S4 says: ice is solid water; warming can melt ice into liquid water. No temperature or speed is supplied.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/18

Retrieval Augmented Generation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read source

S4 supports the material identity and the solid-to-liquid change.



2. Draft two claims

Draft: ice is solid water; warming can turn it into liquid water.



3. Check each claim

Both claims match S4. A claim that melting takes exactly two minutes would not match.

CHANGE ONE THING

The question asks exactly how long this ice takes to melt.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read source	S4 supports the material identity and the solid-to-liquid change.
2. Draft two claims	Draft: ice is solid water; warming can turn it into liquid water.
3. Check each claim	Both claims match S4. A claim that melting takes exactly two minutes would not match.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Both claims match S4. A claim that melting takes exactly two minutes would not match. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

This source-grounding exercise does not establish that every source is complete or scientifically authoritative.

Change one thing in the pictured case

Change one thing

Change: The question asks exactly how long this ice takes to melt.

Predicted result

Prediction: Say that S4 does not provide a time.

Why the result follows

Reason: A general change of state does not determine a duration for an unspecified sample.

Ask while showing the picture

Ask while showing the picture · 10

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Retrieval Augmented Generation, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Note A says library opens at 9. Note B says returns go in the blue box.

2. State the decision to explain

Write the question to explain: Answer opening time with **evidence**.

3. Reveal the worked reasoning

Reveal after **prediction**s: Answer 9 with note A. Retrieval supplies **evidence** and generation phrases it. B does not support opening time.

Hold back the independent task

Keep for an independent attempt: A retrieved passage says the club meets Tuesday; it says nothing about cost. The assistant answers Tuesday, free entry. Which part is grounded and what should replace the unsupported part?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: Approval and capacity are supported by P and Q; lunch lacks **evidence**. A **grounded answer** reports the first two claims and does not invent lunch.

Qualification: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Stage	Teacher action
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Teach the connected idea

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Teach the connected idea

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion practice

Completion task: Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch. Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: Approval and capacity are supported by P and Q; lunch lacks **evidence**. A **grounded answer** reports the first two claims and does not invent lunch.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Building a **grounded answer**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> #### Expected repair
>
> Expected correction: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Chunking and document structure

Target: Explain the **mechanism** and terminology of **chunk**ing and document structure

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: A notice says capacity 20, followed by during maintenance capacity 8. Separate **chunk**s contain each sentence but retrieval returns only the first.

Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: The first **chunk** alone lacks the applicable exception. Retrieve the linked condition and determine maintenance status before asserting capacity 20.

Individual check: For **Chunk**ing and document structure, use its worked case to evaluate this proposed explanation: “The worked result establishes every future case”. Explain the deciding step.

Expected repair

Expected correction: Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Connected teaching cycle: Lexical and vector retrieval

Target: Explain the **mechanism** and terminology of lexical and **vector retrieval**

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: **Query** terms red sensor. D1 text red sensor manual; D2 text red paint. Their lexical scores are 2 and 1. Vector **query** [1, 0], D1=[.4, .9], D2=[.8, .1].

Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: Lexical ranks D1 first; dot product ranks D2 first (.8 versus .4). The supplied task **relevance** favours the sensor manual; vector ranking alone does not prove **relevance**.

Individual check: For Lexical and **vector retrieval**, use its worked case to evaluate this proposed explanation: “The worked result establishes every future case”. Explain the deciding step.

Expected repair

Expected correction: Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

None. Keep the supported state-change explanation and explicitly say the time is not supplied. Changed-case check: Say that S4 does not provide a time.

A general change of state does not determine a duration for an unspecified sample.

Changed-case reasoning: Say that S4 does not provide a time. A general change of state does not determine a duration for an unspecified sample.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Retrieval Augmented Generation investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Run paper RAG with mixed passages, rank **relevance**, draft only from selected **evidence**, cite, and test a missing-answer case.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Check these misconceptions

The worked result establishes every future case: Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Check these misconceptions

The worked result establishes every future case: Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Retrieval Augmented Generation questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Answer 9 with note A. Retrieval supplies **evidence** and generation phrases it. B does not support opening time.

Independent question

Independent question · withhold the key

A retrieved passage says the club meets Tuesday; it says nothing about cost. The assistant answers Tuesday, free entry. Which part is grounded and what should replace the unsupported part?

Expected independent answer

Teacher key · Expected independent answer

Tuesday is supported. Free entry is unsupported; say the passage does not state the cost and seek an appropriate source if needed.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Retrieving one useful passage supports every detail in the final answer.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

It should say the supplied retrieved sources do not establish opening time and request better retrieval. The full collection contains A, so diagnose retrieval separately from generation.

A generator should not use cafe hours as library **evidence**.

Explanations for the additional workbook examples

W1 Explain the mechanism

Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning · W1

Teacher explanation: A **grounded answer** connects each factual claim to **evidence** actually retrieved for the task. Retrieval success does not automatically make generation faithful.

Build a claim-**evidence table** before final wording. Preserve source versions and **qualifier**s, omit unsupported claims or explicitly state the gap, then validate the answer against the table.

W2 Trace the worked case

Workbook question · W2

Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning · W2

Teacher explanation: Approval and capacity are supported by P and Q; lunch lacks **evidence**. A **grounded answer** reports the first two claims and does not invent lunch.

W3 Audit the limit

Workbook question · W3

A report omits this qualification: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning · W3

Teacher explanation: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Workbook question · W4

Source A approves room H; B lists H capacity 7, except maintenance limit 4. Maintenance status is unknown. Write a grounded response.

Teacher reasoning · W4

Teacher explanation: Room H approval is supported. State the conditional capacities 7 normally and 4 during maintenance, with status unresolved; do not choose an unconditional capacity.

W5 Design a revealing follow-up

Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

The new case should test this particular **mechanism** rather than merely restate the original result.

Chunking and document structure — W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning

Teacher explanation: **Chunk**ing divides documents into retrievable units. A useful **chunk** retains enough structure to interpret a passage, including headers and exceptions.

Give each **chunk** its document/version ID, section and **offset**s. Preserve **qualifier**s or link adjacent material. **Overlap** can repeat information; repeated **chunk**s are not independent sources.

Chunking and document structure — W2 Trace the worked case

A notice says capacity 20, followed by during maintenance capacity 8. Separate **chunk**s contain each sentence but retrieval returns only the first. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning

Teacher explanation: The first **chunk** alone lacks the applicable exception. Retrieve the linked condition and determine maintenance status before asserting capacity 20.

Chunking and document structure — W3 Audit the limit

A report omits this qualification: Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning

Teacher explanation: Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

Chunking and document structure — W4 Apply a new case independently

A rule section says rooms close at 17:00; an exception paragraph says Friday closes at 16:00. The question is Friday closing. Design a complete **chunk** reference.

Teacher reasoning

Teacher explanation: Return both rule and linked Friday exception with document/version and section references; answer 16:00 from the exception. An isolated 17:00 **chunk** is insufficient.

Chunking and document structure — W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

The new case should test this particular **mechanism** rather than merely restate the original result.

Lexical and vector retrieval — W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning

Teacher explanation: Lexical retrieval matches terms; **vector retrieval** compares numerical representations. Each can fail when the representation misses the relevant distinction.

For a toy lexical score, count distinct **query** terms present after lowercase exact-word matching. For a toy vector score use a stated dot product. Compare **relevance** using independent judgments, not score magnitude alone.

Lexical and vector retrieval — W2 Trace the worked case

Query terms red sensor. D1 text red sensor manual; D2 text red paint. Their lexical scores are 2 and 1. Vector **query** [1, 0], D1=[.4, .9], D2=[.8, .1]. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning

Teacher explanation: Lexical ranks D1 first; dot product ranks D2 first (.8 versus .4). The supplied task **relevance** favours the sensor manual; vector ranking alone does not prove **relevance**.

Lexical and vector retrieval — W3 Audit the limit

A report omits this qualification: Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning

Teacher explanation: Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

Lexical and vector retrieval — W4 Apply a new case independently

Query terms blue battery. E1 text blue battery guide, vector [.2, .8]; E2 text blue ribbon, vector [.9, .1]. **Query** vector [1, 0]. Compute both rankings.

Teacher reasoning

Teacher explanation: Lexical E1=2, E2=1; vector E1=.2, E2=.9 so rankings disagree. Check **relevance** for the battery task; do not equate dot product with source truth.

Lexical and vector retrieval — W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

The new case should test this particular **mechanism** rather than merely restate the original result.

Assess independent transfer

Independent question · withhold the key

Selected case: Source A approves room H; B lists H capacity 7, except maintenance limit 4. Maintenance status is unknown. Write a grounded response.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: Room H approval is supported. State the conditional capacities 7 normally and 4 during maintenance, with status unresolved; do not choose an unconditional capacity.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

TEACHER GUIDE · 6 OF 9

Connect and extend learning

> Connect Retrieval Augmented Generation to Verified Study Buddy, adapt support and plan a later changed-case check.

Verified Study Buddy

Implement question → retrieve → inspect → answer → cite → verify as the core architecture.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 19: User, Need, and AI Contribution. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Room H approval is supported. State the conditional capacities 7 normally and 4 during maintenance, with status unresolved; do not choose an unconditional capacity. Require intermediate reasoning and the stated scope.

Fresh retry: Source approves J; register says capacity 9 unless exam setup limits it to 6. Setup unknown.

Expected reasoning: Report approval and both capacity conditions; ask which setup applies before giving a single applicable limit.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Claim support and citation evaluation. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Readiness and reassessment for Chunking and document structure

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Return both rule and linked Friday exception with document/version and section references; answer 16:00 from the exception. An isolated 17:00 **chunk** is insufficient. Require intermediate reasoning and the stated scope.

Fresh retry: General limit 30; exam-week exception 18. Question concerns exam week. Expected reasoning: Preserve both scope and exception; applicable limit 18. Two **overlap**ping **chunk**s do not count as two corroborating sources.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Readiness and reassessment for Lexical and vector retrieval

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Lexical E1=2, E2=1; vector E1=.2, E2=.9 so rankings disagree. Check **relevance** for the battery task; do not equate dot product with source truth. Require intermediate reasoning and the stated scope.

Fresh retry: **Query** green motor; F1 contains both, vector [.3, .7]; F2 green fabric, vector [.6, .4]; **query** [1, 0]. Expected reasoning: Lexical F1=2, F2=1; dot products .3, .6 favour F2.

Use **relevance judgment**s to evaluate the representation. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Plan the delayed return

Suggested return points in this level: ai-3/19, ai-3/21, ai-3/24. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

TEACHER GUIDE · 7 OF 9

Prepare, teach and assess

> Use this preparation summary alongside the detailed Retrieval Augmented Generation lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain the **mechanism** and terminology of building a **grounded answer**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch. Show the intermediate steps and give the conclusion with its scope.

Supported practice answer

Reasoned practice answer: Teacher explanation: Approval and capacity are supported by P and Q; lunch lacks **evidence**. A **grounded answer** reports the first two claims and does not invent lunch.

Check every learner before discussion: Use the worked case for Building a **grounded answer**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The question asks exactly how long this ice takes to melt. Require a **prediction** and a reason before comparing. Expected result: Say that S4 does not provide a time.

Reason: A general change of state does not determine a duration for an unspecified sample.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

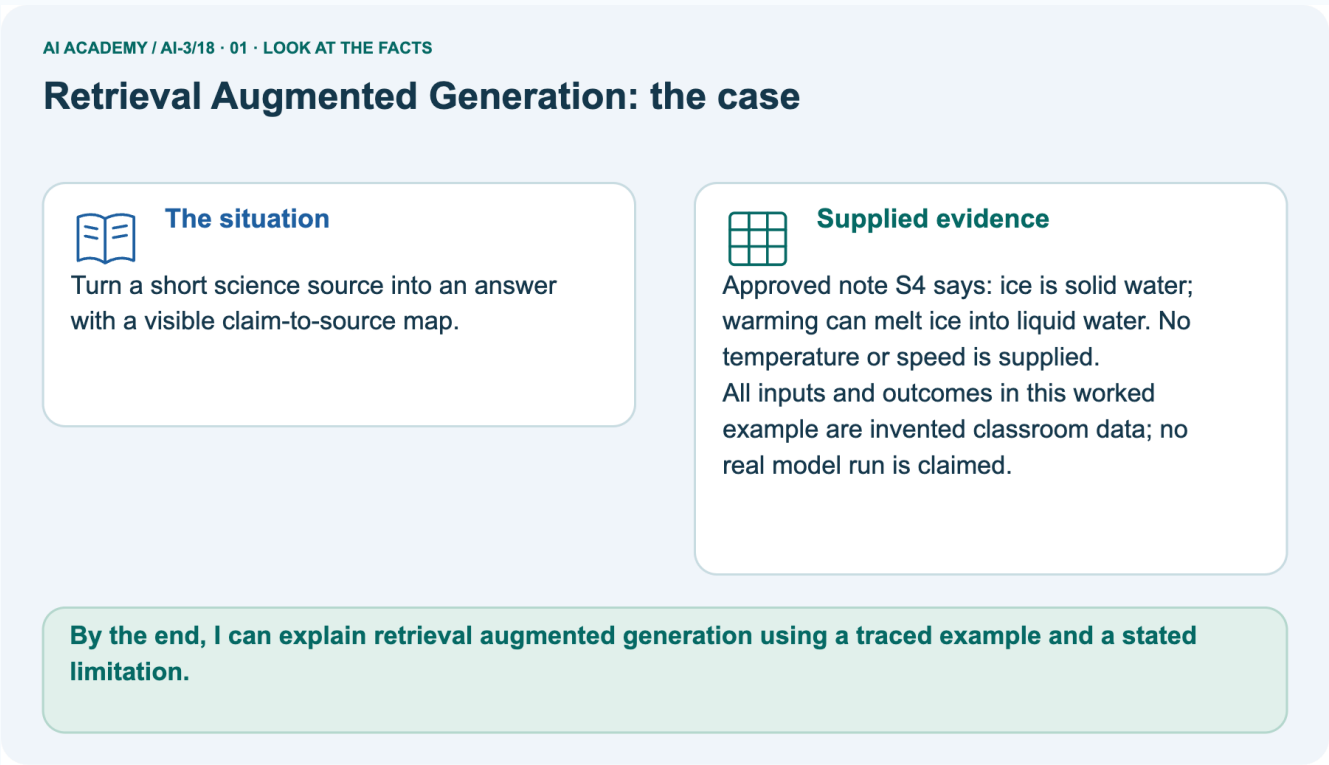
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Retrieval Augmented Generation graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Retrieval Augmented Generation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Turn a short science source into an answer with a visible claim-to-source map. Supplied facts: Approved note S4 says: ice is solid water; warming can melt ice into liquid water. No temperature or speed is supplied.

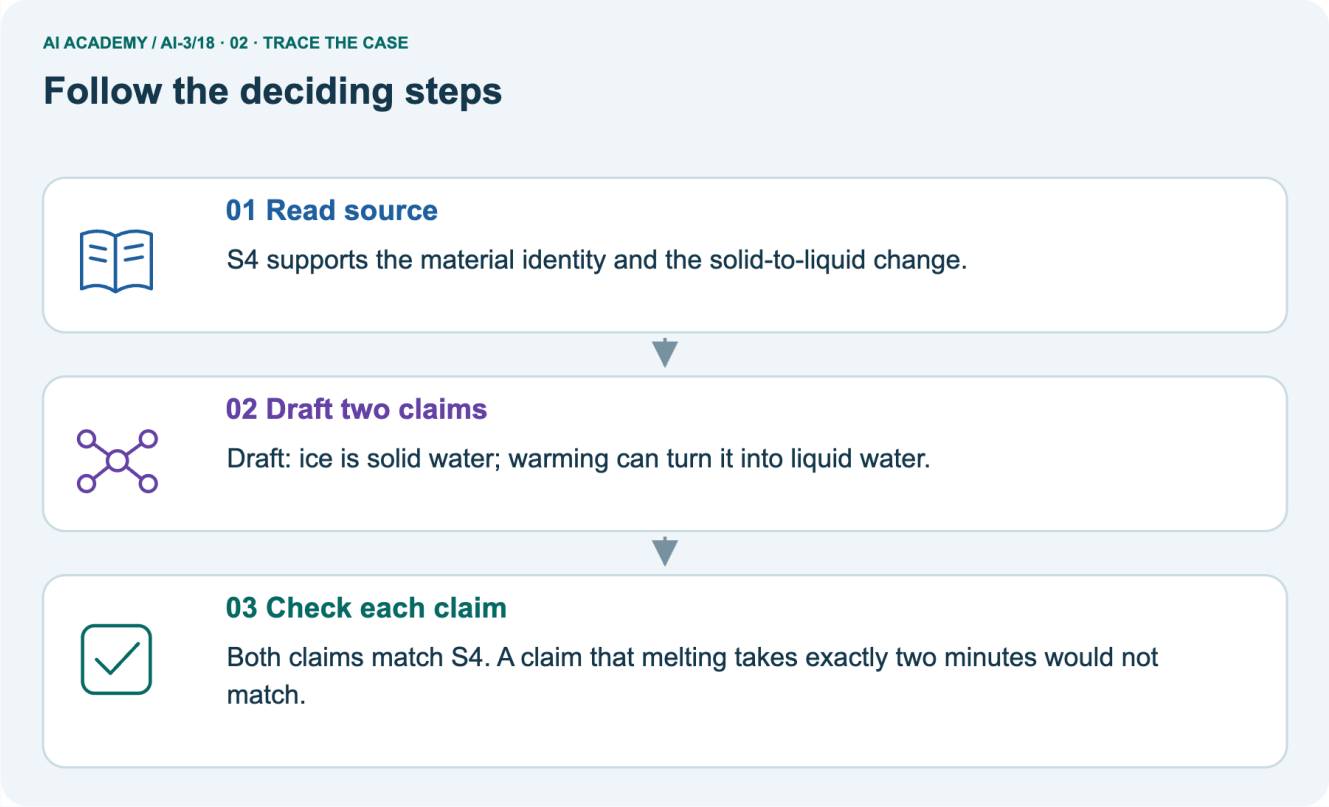
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read source: S4 supports the material identity and the solid-to-liquid change. → Draft two claims: Draft: ice is solid water; warming can turn it into liquid water. → Check each claim: Both claims match S4.

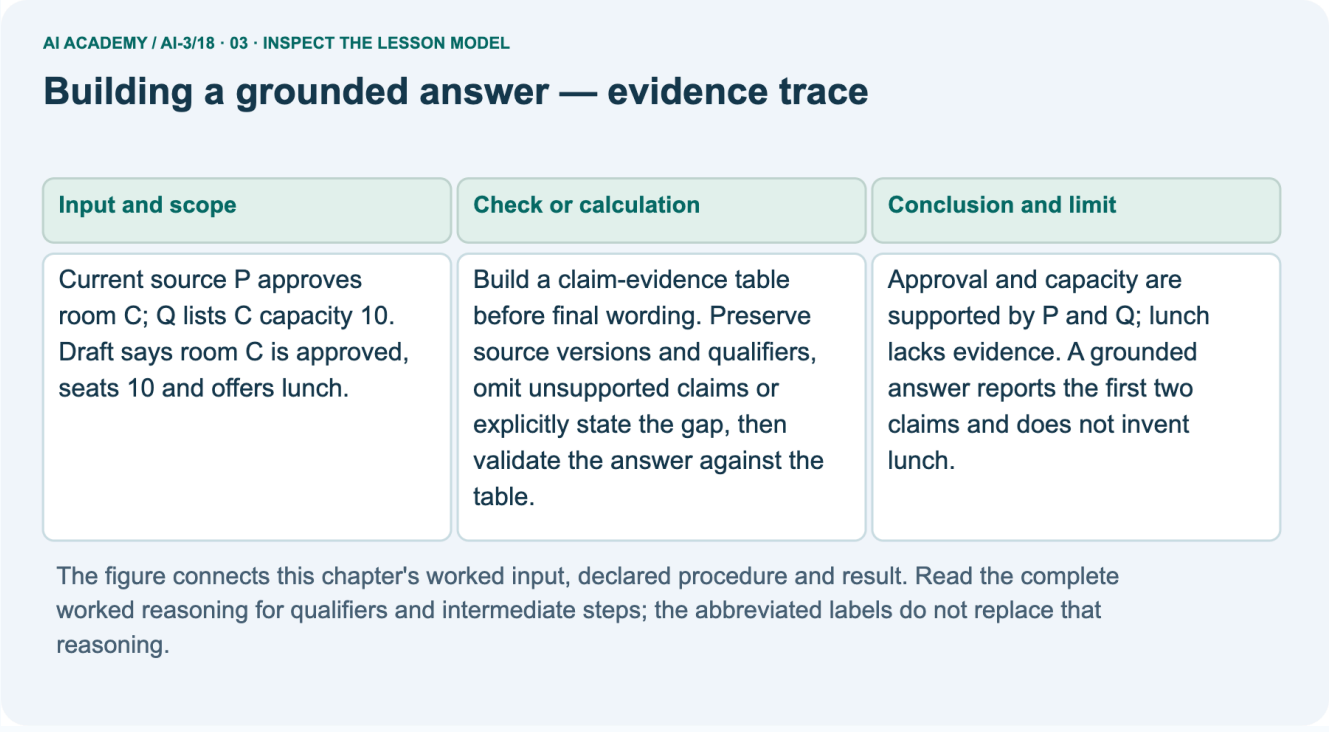
A claim that melting takes exactly two minutes would not match.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Building a grounded answer — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Building a **grounded answer** — **evidence** trace. Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch. | Build a claim-**evidence table** before final wording.

Preserve source versions and **qualifier**s, omit unsupported claims or explicitly state the gap, then validate the answer against the table. | Approval and capacity are supported by P and Q; lunch lacks **evidence**.

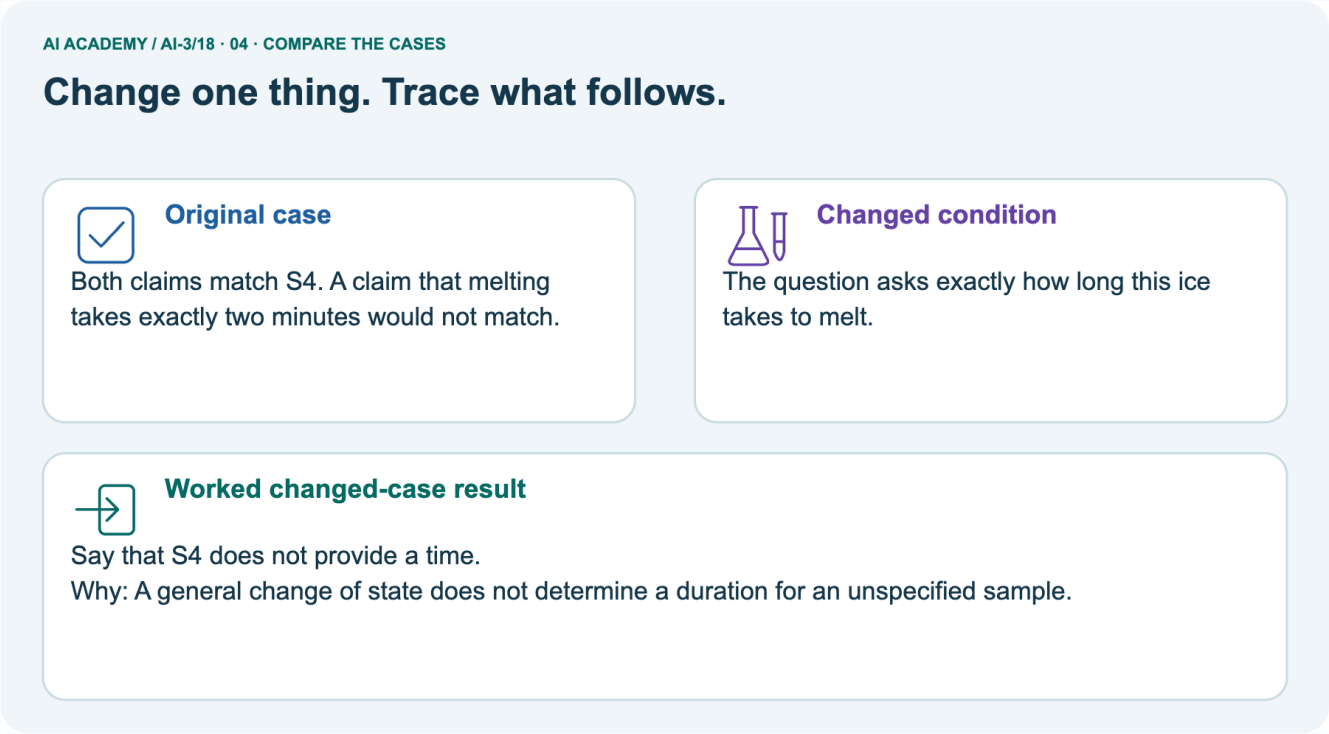
A **grounded answer** reports the first two claims and does not invent lunch.

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both claims match S4. A claim that melting takes exactly two minutes would not match. Changed condition: The question asks exactly how long this ice takes to melt.

Expected result: Say that S4 does not provide a time. Reason: A general change of state does not determine a duration for an unspecified sample.

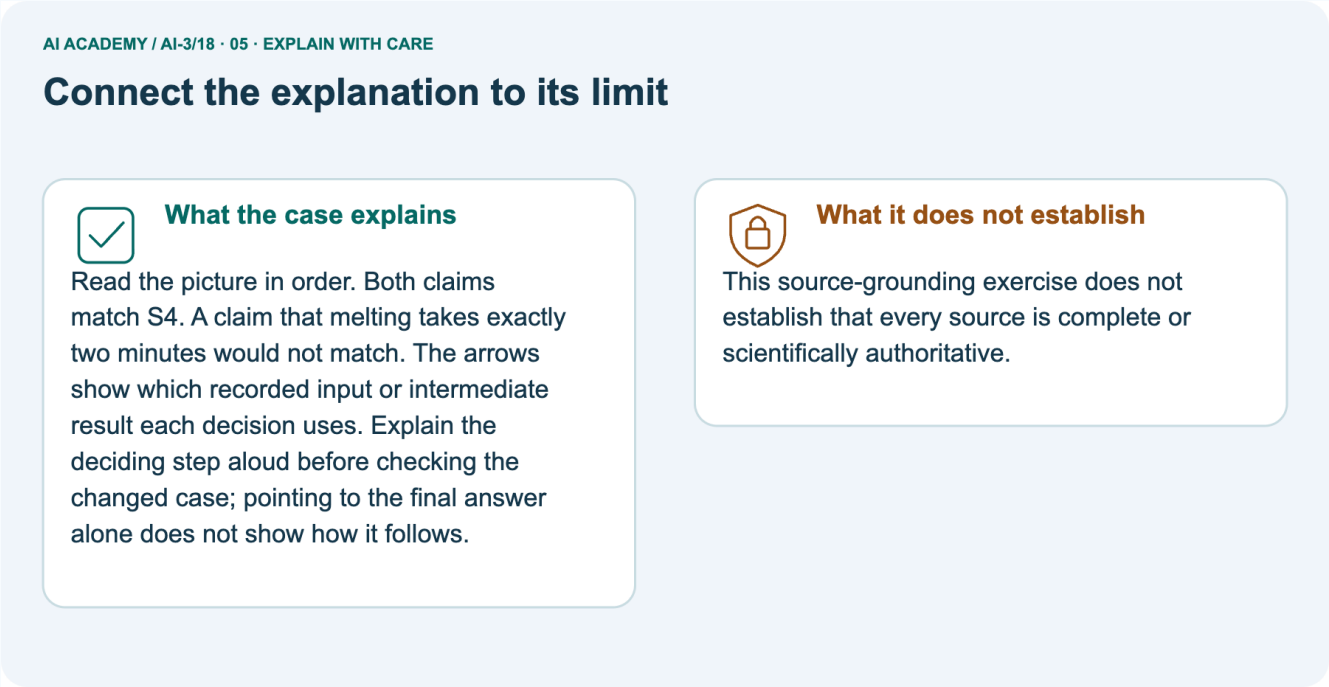
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A general change of state does not determine a duration for an unspecified sample.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Both claims match S4. A claim that melting takes exactly two minutes would not match. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: This source-grounding exercise does not establish that every source is complete or scientifically authoritative.

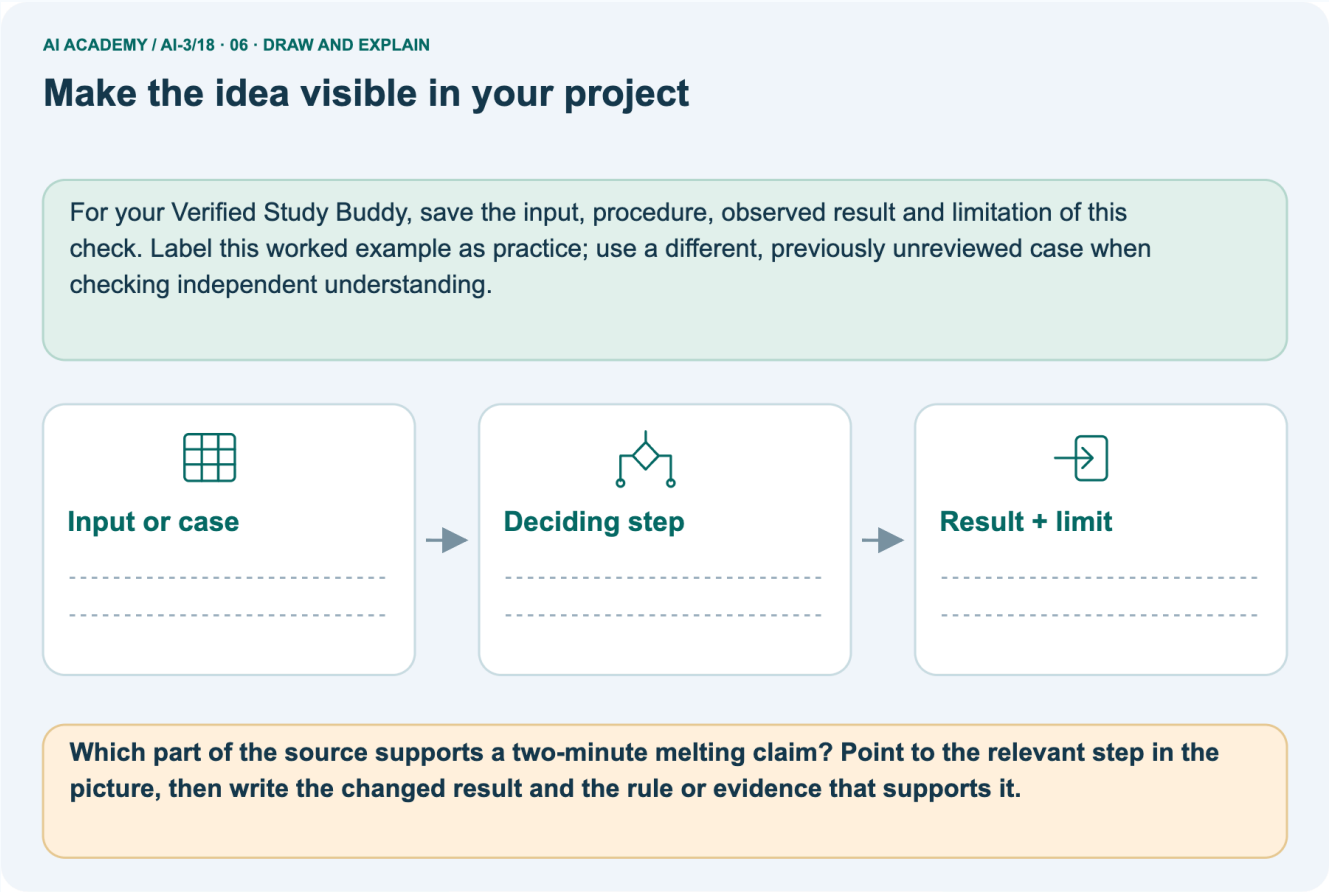
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This source-grounding exercise does not establish that every source is complete or scientifically authoritative.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: None. Keep the supported state-change explanation and explicitly say the time is not supplied. Changed-case check: Say that S4 does not provide a time.

A general change of state does not determine a duration for an unspecified sample.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.